

n	$p(z)$	$g(z)$	a, b	$N_{\frac{z^4}{p(z)}}$
4	$z^4 - 1$	-	-	$\frac{z(z^4+3)}{4}$
3	$(z-1)^2(z^2+az+b)$	$(a-2)z^2+(-3a+2b)z-4b$	$a=2$ $b=3$	$\frac{z(z^3+z^2+z+9)}{12}$
2	(i) $(z-1)^2(z-a)^2$	$2(2a-(a+1)z)$	$a=-1$	$\frac{z(z^2+3)}{4}$
2	(ii) $(z-1)^3(z-a)$	$-(a+3)z+4a$	$a=-3$	$\frac{z(z^2+2z+9)}{12}$
1	$(z-1)^4$	4	-	$\frac{z(z+3)}{4}$

Table 1: Newton maps $N_{\frac{z^4}{p(z)}}$ with an exceptional point

1. Let p be generic. If p is linear, then it follows from the Scaling property that $N_R(z)$ is conjugate to $N_{\frac{z}{z-1}}$.

Let p be non-linear. As p is generic and $N_{\frac{z^d}{p(z)}}$ is a polynomial, we have $zp'(z) - dp(z) = \alpha$ for some non-zero α . Letting $y = p(z)$, we have the first-order linear differential equation $y' - \frac{d}{z}y = \frac{\alpha}{z}$. The solution is

$$y \cdot \frac{1}{z^d} = \int \frac{\alpha}{z} \cdot \frac{1}{z^d} dz + \beta = -\frac{\alpha}{dz^d} + \beta,$$

for an arbitrary constant β . Therefore, $p(z) = \beta z^d - \frac{\alpha}{d}$ and $R(z) = \frac{z^d}{\beta z^d - \frac{\alpha}{d}}$. Consider c such that $c^d = \frac{\alpha}{d\beta}$ and use the Scaling property to see that N_R is conjugate to $N_{\beta R(cz)}$. We are done since $\beta R(cz) = \frac{z^d}{z^d-1}$ and the resulting Newton map is $\frac{z^{d+1}+(d-1)z}{d}$.

2. Let n denote the number of distinct roots of p . Using the Scaling property, we assume without loss of generality that 1 is a multiple root of p whenever p is not generic. Along with this, what is going to be repeatedly used in all the following cases is that $g(z)$ is a non-zero constant (see Equation (3.1)).
 - (a) Let $\deg(p) = 3$. Then there are three cases depending on the values of n . If $n = 3$, then p is generic, and from the first part of this theorem, it follows that $p(z) = z^3 - 1$, and hence $N_R(z) = \frac{1}{3}z(z^3 + 2)$. If $n = 2$, then p has a root with multiplicity 2 and therefore $p(z) = (z-1)^2(z-a)$, where $a \neq 0, 1$. In this case, $g(z) = -(a+2)z + 3a$ and therefore $a = -2$. Thus, $N_R(z) = \frac{z}{6}(z^2 + z + 4)$. If $n = 1$, then $p(z) = (z-1)^3$, and we get $N_R(z) = \frac{1}{3}z(z+2)$.
 - (b) Let $\deg(p) = 4$. All possible cases of p and the resulting Newton maps are given in Table 1.
 - (c) For $\deg(p) = 5$, all the possible forms of p and resulting Newton maps are given in Table 2.